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Optimization of Prehospital Triage of Patients With Suspected Ischemic Stroke

Results of a Mathematical Model

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Background and Purpose—Prehospital routing algorithms for patients with suspected stroke because of large vessel occlusions should account for likelihood of benefit from endovascular therapy (EVT), risk of alteplase delays, and transport times. We built a mathematical model to give a real-time, location-based optimal emergency medical service routing location based on local resources, transport times, and patient characteristics.

Methods—Using location, onset time, age, sex, and prehospital stroke severity, we calculated odds of a favorable outcome for a patient with suspected large vessel occlusions under 2 scenarios: direct to EVT-capable hospital versus transport to the nearest alteplase-capable hospital with transfer to EVT-capable hospital if appropriate. We project lifetime outcomes incorporating disability, quality of life utility, and cost. Multiple parameter sets of center-specific times (eg, door to alteplase) were randomly selected within a clinically plausible range to account for the model sensitivity to these estimates; for each iteration, the optimal strategy was defined as the most cost-effective outcome (threshold, $100 000 per quality-adjusted life-years gained). After 1000 simulations, the most frequently occurring optimal strategy was the final recommendation, with its strength measured as the proportion of runs for which it was optimal.

Results—Routing recommendations were highly sensitive to small changes in model input parameters. Under many scenarios, the recommendations for direct transfer to the EVT site increased with increasing stroke severity and geographic proximity but did not vary substantially with respect to sex, age, or onset time.

Conclusions—We present a mathematical decision model that determines ideal prehospital routing recommendations for patients with suspected stroke because of large vessel occlusions, with consideration of patient characteristics and location at onset. This model may be further refined by incorporating real-time data on traffic patterns and actual EVT and alteplase timeliness performance. Further studies are needed to verify model predictions. **(*Stroke*. 2018;49:2532-2535. DOI: 10.1161/STROKEAHA.118.022041.)**

Key Words: emergency medical services ■ endovascular procedures ■ models, theoretical ■ stroke ■ triage

In the prehospital setting, emergency medical services (EMS) personnel typically transport patients with suspected stroke to the closest alteplase-capable center (acute stroke-ready hospital or primary stroke center [PSC]), most of which are not endovascular thrombectomy-capable centers (comprehensive stroke center [CSC]). Yet some patients may benefit from routing past a nearby PSC directly to a CSC, at the potential cost of delayed alteplase administration.

Previous work has found that routing recommendations are highly sensitive to transport and door-to-needle (DTN) times.[1,2] However, those studies assumed presence of acute ischemic stroke although, in practice, a patient's stroke diagnosis is unknown.[3] Furthermore, although prior models predicted likelihood of good outcome, the clinical difference may be negligible. Our primary objective was to build a mathematical model that simulates hypothetical patients with stroke-like symptoms in the prehospital setting to determine the optimal prehospital EMS routing strategy based on long-term outcomes, incorporating real-time transport times and geographical location.

Methods

The authors declare that all supporting data, including model source code, are available within the article (and its online-only Data Supplement).

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Table 1. Hypothetical Stroke Patient Located in 3 US Locations for Illustrative Purposes

Location	Transport Scenarios as of February 6, 2018	Age, y	Sex	Time From Onset at EMS Scene Arrival, min	Stroke Severity, RACE (NIHSS)
Kingston, IL	Option 1: 31 min to PSC with 62-min transfer to CSC	70	F	30 vs 70	3 (6.8) vs 5 (11.6)
	Option 2: 53 min direct to CSC				
	PSC: OSF Saint Anthony Medical Center in Rockford, IL				
	CSC: Alexian Brothers in Elk Grove Village, IL				
Brookfield, CT	Option 1: 14 min to PSC with 60-min transfer to CSC	80	M	45 vs 90	4 (9.2) vs 6 (14.0)
	Option 2: 59 min direct to CSC				
	PSC: Danbury Hospital in Danbury, CT				
	CSC: Hartford Hospital in Hartford, CT				
Beaver Dam, AZ	Option 1: 37 min to PSC with 117-min transfer to CSC	65	F	10 vs 70	3 (6.8) vs 8 (18.7)
	Option 2: 91 min direct to CSC				
	PSC: Dixie Regional Medical Center in St George, UT				
	CSC: Sunrise Hospital and Medical Center in Las Vegas, NV				

CSC indicates comprehensive stroke center; EMS, emergency medical service; F, female; M, male; NIHSS, National Institutes of Health Stroke Scale; PSC, primary stroke center; and RACE, rapid arterial occlusion evaluation.

We constructed a mathematical model to determine the optimal transport decision for prehospital routing of patients with suspected stroke. The simulation starting point is at the scene when the EMS provider first reaches the patient with the presumed stroke. Inputs are patient characteristics measurable by EMS providers and geographical location to inform real-time transit times to nearby PSC and CSC. PSCs and CSCs are modeled with separate distributions for DTN times, which affects all patients with ischemic stroke. Prehospital stroke severity is measured using the rapid arterial occlusion evaluation score which is strongly correlated with the National Institutes of Health Stroke Scale[4] and probability of large vessel occlusions.[5]

The model output is a recommendation for initial transport to a PSC or CSC. The 2 primary outcomes simulated per strategy are lifetime cost and quality-adjusted life-years. Scenarios are compared using a cost-effectiveness approach with a maximum threshold of $100000 per quality-adjusted life-years gained. To account for the model's sensitivity to center-specific performance (eg, DTN times), each scenario is run multiple times with randomly chosen time values within a plausible distribution; the final recommendation is the most frequently occurring optimal destination. The model is written in Python 3.6.1 (online-only Data Supplement).[6]

Because the value of this model is in its application to specific patient scenarios, we selected a few varied hypothetical but realistic clinical situations in which the model might be used. For each vignette, we identified an approximate location for the scene of the stroke and estimated travel times to stroke centers (PSCs and CSCs) identified with Joint Commission certification data. Vignette inputs are summarized in Table 1. Details about vignette map construction are provided in the online-only Data Supplement.

We did not obtain Institutional Review Board approval because no individual patient data were used.

Results

Our mathematical model provides real-time, location-based, optimal EMS routing destinations for patients with suspected stroke based on local resources, transport times, and patient

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Table 2. Vignette Results

Location	Time From Onset, min	Stroke Severity (RACE)	Optimal Destination	Distribution	
				%CSC	%PSC
IL	30	3	CSC	58.8 (55.7–61.9)	41.2 (38.1–44.3)
		5	CSC	84.9 (82.6–87.2)	15.1 (12.8–17.4)
	70	3	CSC	54.2 (51.1–57.4)	45.8 (42.6–48.9)
		5	CSC	81.9 (79.5–84.3)	18.1 (15.7–20.5)
CT	45	4	PSC	4.9 (3.5–6.3)	95.1 (93.7–96.5)
		6	PSC	23.2 (20.6–25.9)	76.8 (74.1–79.4)
	90	4	PSC	3.6 (2.5–4.8)	96.4 (95.2–97.5)
		6	PSC	18.1 (15.8–20.5)	81.9 (79.5–84.2)
AZ	10	3	PSC	12.0 (10.0–13.9)	88.0 (86.1–90.0)
		8	CSC	99.6 (99.2–100.0)	0.4 (0.0–0.8)
	70	3	PSC	15.0 (12.8–17.1)	85.0 (82.9–87.2)
		8	CSC	99.4 (98.9–99.9)	0.6 (0.1–1.1)

CSC indicates comprehensive stroke center; PSC, primary stroke center; and RACE, rapid arterial occlusion evaluation.

characteristics. Overall, we found that routing recommendations were sensitive to small changes in input parameters and, in general, that severe strokes are more frequently routed to CSCs. Sites were chosen for these illustrative vignettes based on varied local resources. Results are summarized in Table 2, with graphical illustrations in the Figure (AZ) and Figures I (IL) and II (CT) in the online-only Data Supplement.

Discussion

We report the development of a novel, comprehensive mathematical model that builds on prior published work with the addition of critical prehospital elements and economic and probabilistic outcome modeling. This more nuanced approach to the question of optimal prehospital EMS routing reflects a realistic approach to the real-world challenges encountered by EMS personnel. The model also allows for further refinement with actual hospital-level performance if these data become available. The freely available open-source platform of the model also enables further use for academic and on-the-ground applications, for example in refining stroke systems of care that are fine-tuned to local environments.

This model incorporates local resources (proximity of PSCs and CSCs), transport times based on real-time traffic data, and patient characteristics, including time of onset and severity of symptoms. Because our model is designed to give specific and actionable outputs for any given input scenario, it is specifically not intended to make broad generalizations about destination routing plans because the model clearly shows that the outputs are exquisitely sensitive to minor variations in the input parameters. An important finding from our model is that small changes in time from symptom onset or distance to a nearby CSC lead to significant changes in routing recommendations, as illustrated by the case vignettes. We presented the model outputs as specific case vignettes to demonstrate the ease of personalization of recommendations compared with simple uniform rules that are designed to be one size fits all.

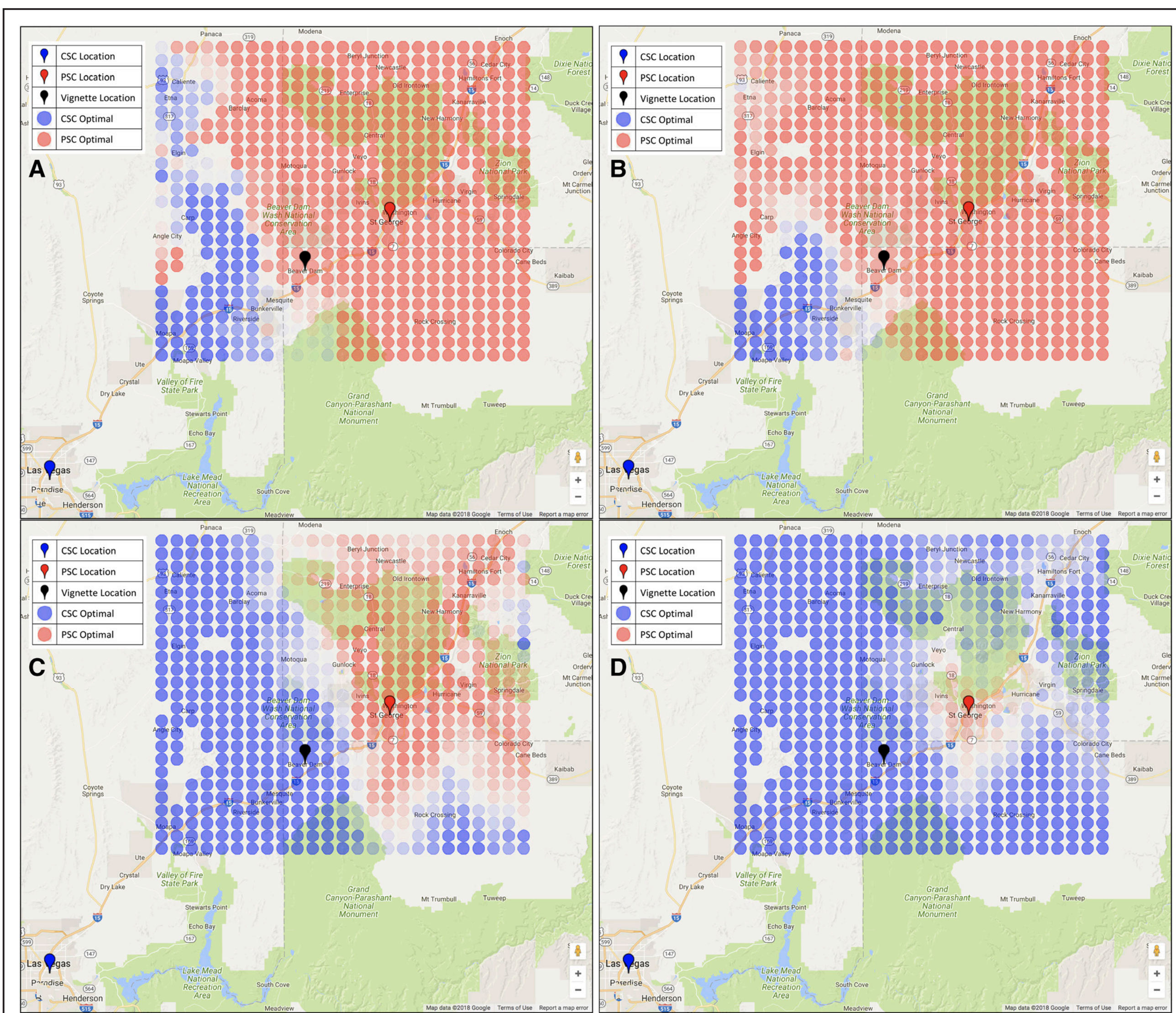

Figure. Optimal strategies for a 65-y-old woman 10 min from symptom onset with rapid arterial occlusion evaluation (RACE) 3 (**A**) and RACE 8 (**C**) and 70 min from symptom onset with RACE 3 (**B**) and RACE 8 (**D**). Red circles indicate primary stroke center (PSC) is optimal, and blue circles indicate comprehensive stroke center (CSC) is optimal. More lightly shaded locations indicate less certainty in the optimal destination. Red markers show primary stroke centers, blue markers show comprehensive stroke centers, and the black marker indicates Beaver Dam—the vignette location. Map data are provided by Google.

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Our model has limitations, including reliance on sparse data for long-term outcomes, not accounting for emergency department crowding and the use of a single PSC and CSC in our algorithm. These limitations are further detailed in the online-only Data Supplement.

Our results are similar to prior work by Holodinsky et al,[2] who reported that transport decisions for patients with acute ischemic stroke were highly sensitive to transportation times between centers and to DTN times. Our version of the model adds these 4 additional key features: (1) it does not assume that acute ischemic stroke status can be determined with certainty in the field, (2) it incorporates the diagnostic uncertainty at the time of EMS transport decision-making and assigns probabilities of hemorrhagic stroke and stroke mimic diagnoses, (3) in contrast to previous studies focusing on good-outcome probabilities, we used quality-adjusted life-years and costs to quantify differences between transport strategies and to provide a clear clinical interpretation to an otherwise mathematical and probabilistic answer, and (4) the open-source and dynamic nature of the model lends itself to incorporation of new data as it becomes available, such as institutional DTN or door-to-puncture times.

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Disclosures

Dr Schwamm reports serving as Stroke Systems of Care consultant to the Massachusetts Department of Public Health (modest), as chair of the American Heart Association Hospital Accreditation Science Subcommittee and cochair of Mission-Lifeline Stroke (unpaid), and as a consultant to Penumbra (significant) and Medtronic (significant). In addition, Dr Schwamm has received significant grants from the National Institute of Neurological Disorders and Stroke and Genentech. Dr Hur reports equity (modest) in Atlas 5D. The other authors report no conflicts.

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